

Problem 1

$$\lim_{x \rightarrow 0} \frac{\sin x}{x}$$

Solution. $\lim_{x \rightarrow 0} \frac{\sin(x)}{x} = 1$

Answer:

1

Problem 2

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 5 & 6 & 0 \end{pmatrix} \quad \det(A)$$

Solution. $A \ 3 \times 3$

/ $\det 1$ **Answer:**

1

Problem 3

$$x + y + z = 6 \quad 2x - y + z = 3 \quad x + 2y - z = 2$$

 $\therefore \textcolor{red}{LLM}$

Problem 4

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$$

Solution. $|I \ A| = 0$

$$\begin{matrix} 1 & 1 \\ 1 & 1 \end{matrix} \begin{bmatrix} -1 \\ 1 \end{bmatrix} \begin{matrix} 3 \\ 1 \end{matrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Answer:

$11 \begin{bmatrix} -1 \\ 1 \end{bmatrix}; 31 \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

Problem 5

$$y'' - 5y' + 6y = 0$$

Solution. $26y(x) - 5\frac{d}{dx}y(x) + \frac{d^2}{dx^2}y(x) = 0$

dsolve

$$y(x) = (C_1 + C_2 e^x) e^{2x}$$

Answer:

$$y(x) = (C_1 + C_2 e^x) e^{2x}$$

Problem 6

$$g = 9.8 \text{ m/s}^2$$

Solution. $v = gt = \frac{1}{2}gt^2 = 9.8 \text{ m/s}^2$

$$t = 9.8 \text{ s} \quad v = 9.8 \times 9.8 = 96.04000000000002 \text{ m/s}$$

$$s = \frac{1}{2} \times 9.8 \times 9.8^2 = 470.5960000000001 \text{ m}$$

Answer:

$$v = 96.04000000000002 \text{ m/s} = 470.5960000000001 \text{ m}$$

Auto-generated: 5/6 problems solved by SymPy.